Evaluating Federated Object Detection for Connected Autonomous Vehicles

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Introduction

Object detection is crucial for Connected Autonomous Vehicles (CAVs) to perceive their surroundings. Centralized training of models often achieves promising accuracy, fast convergence, and simplified training process, falls short in scalability, adaptability, and privacy preserving. Federated learning (FL), by contrast, enables collaborative, privacy-preserving, and continuous training across naturally distributed CAVs shown in Fig 1.

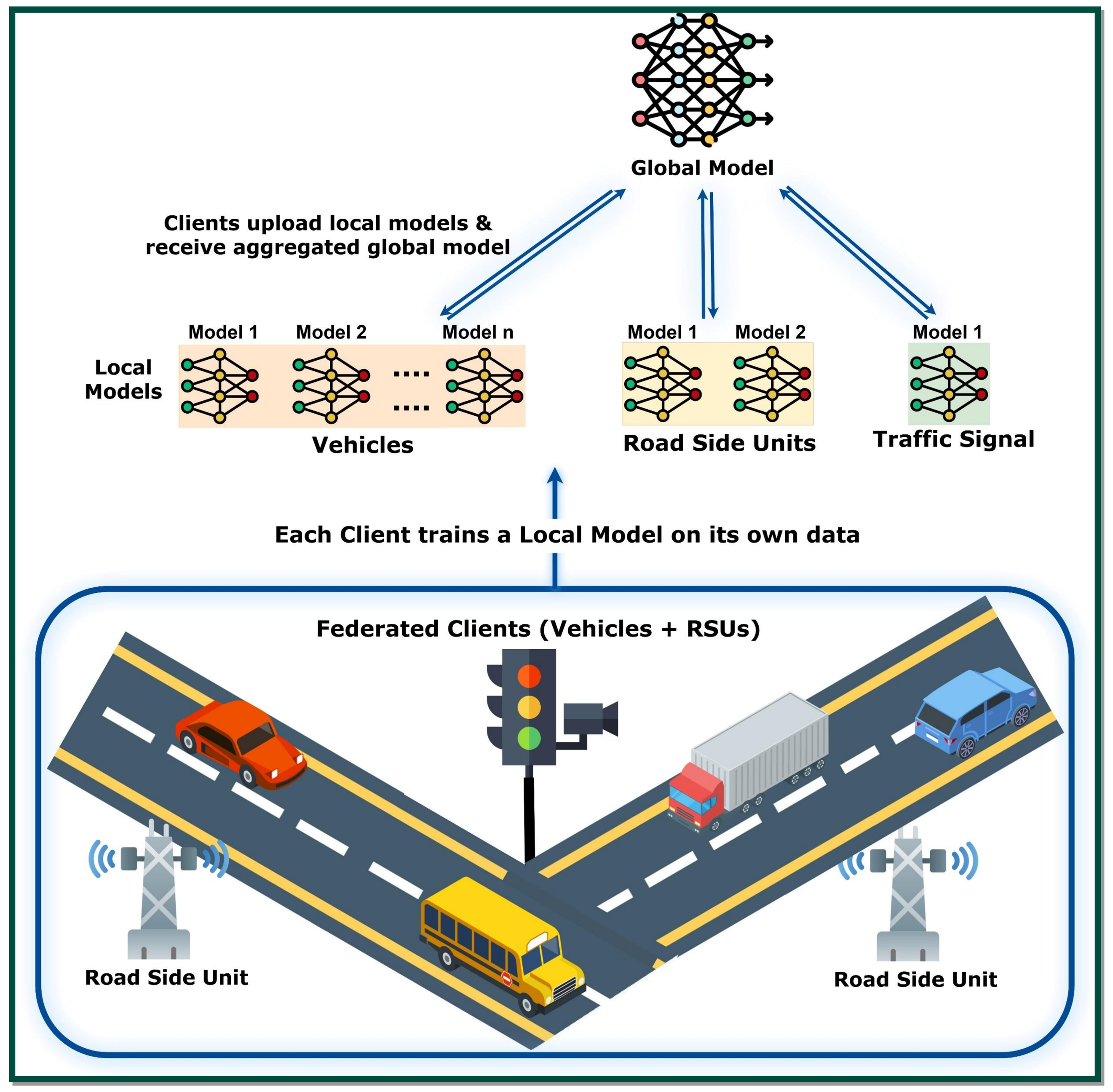

Figure 1. Federated Learning for CAVs

Motivation

Current FL studies focus on accuracy but overlook feasibility. We evaluate models under diverse conditions and resource limits to assess the real-world readiness of FL for CAV deployment. Practical deployment must address three critical factors:

(i) Heterogeneity from non-IID data distributions,
(ii) Constrained onboard computing hardware
(iii) Environmental variability such as lighting and weather

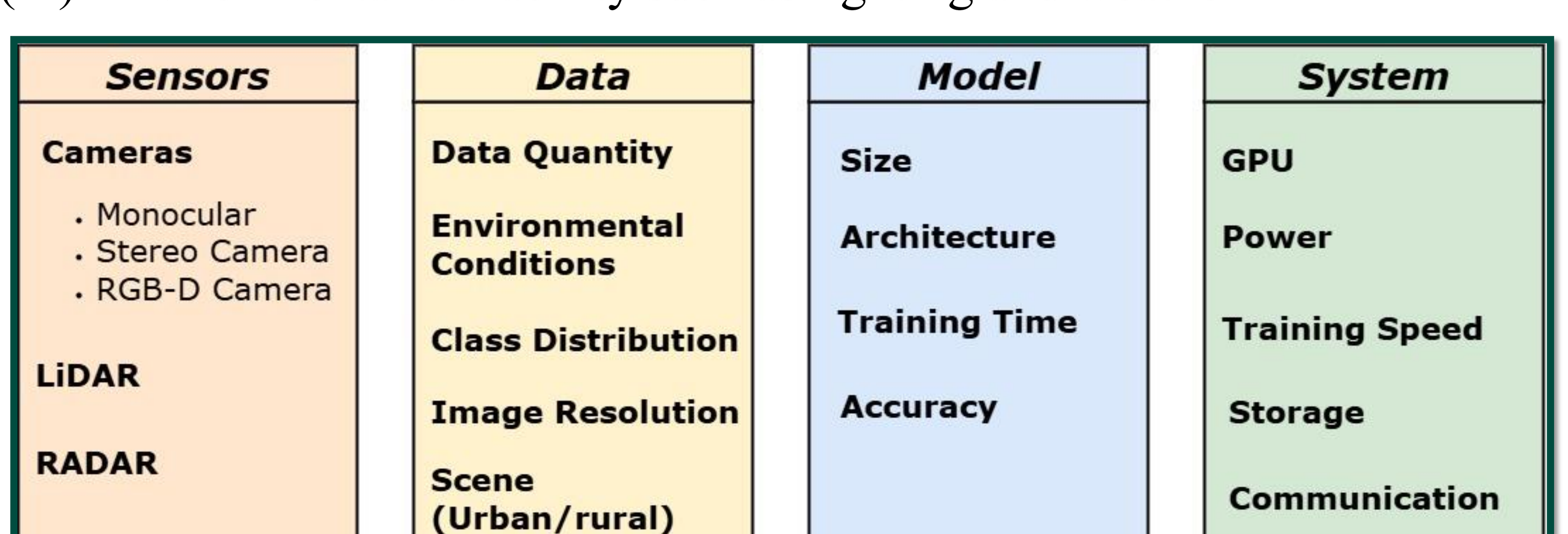

Figure 2. Heterogeneity in Connected Autonomous Vehicles

FL Evaluation Framework

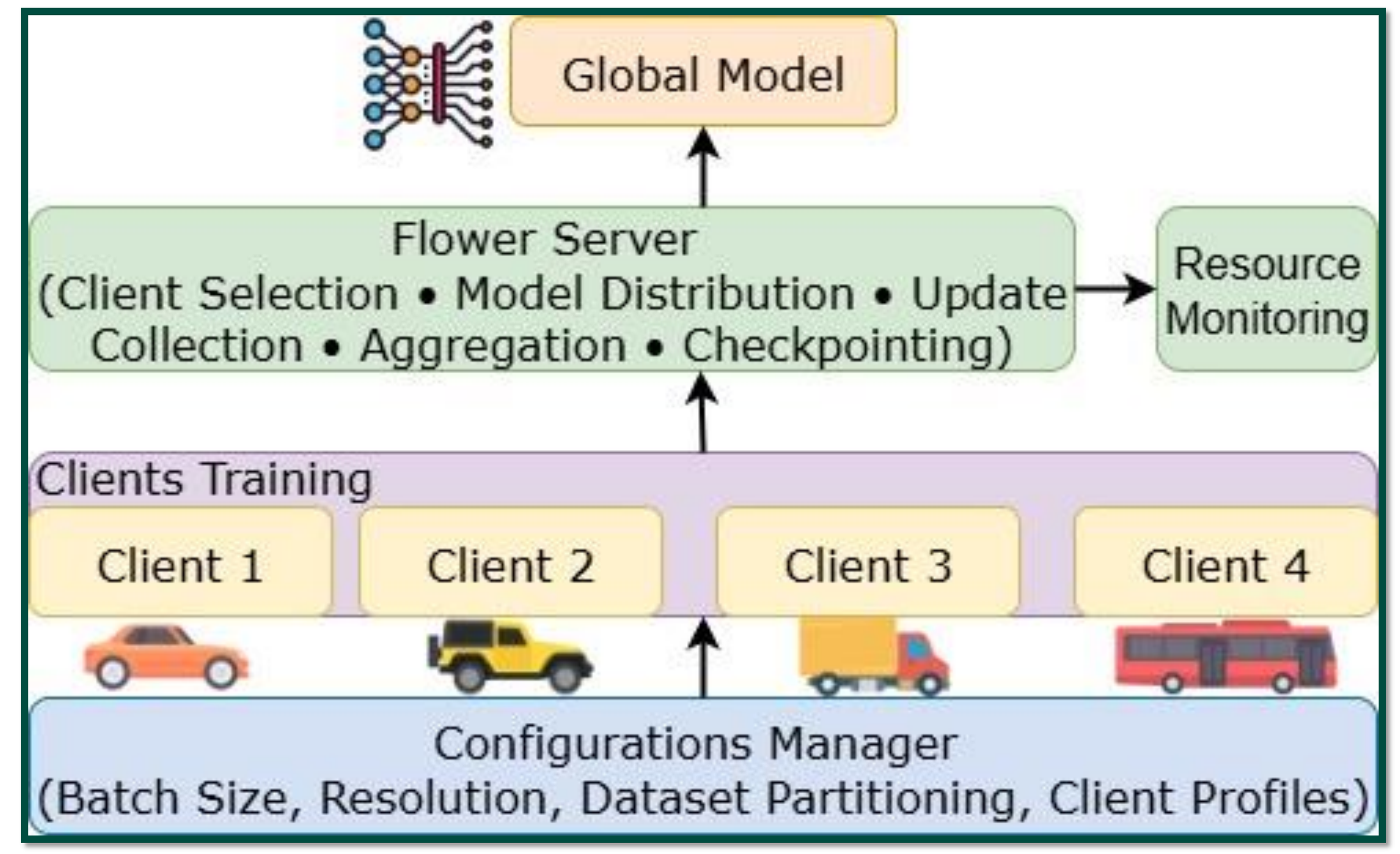

Figure 3. FL Setup with Flower for Object Detection Evaluation

Open Research Problems

Datasets: Limited diversity, static snapshots, and lack of standardized multimodal benchmarks hinder reproducibility and deployment.

Models: Detection struggles with small objects, long-tail classes, and robustness; federated-ready, secure, and efficient designs are needed.

Aggregation: Adaptive strategies must address fairness, resource variability, bandwidth, and fault tolerance under dynamic conditions.

Security & Privacy: Defenses against poisoning, backdoors, and inference leakage remain unresolved; lightweight privacy-preserving and unlearning methods are essential.

Environment: Weather and lighting shifts degrade generalization; robust multimodal adaptation and standardized evaluation protocols are lacking.

Experiment Results

Object Detection Models: YOLOv5, v8, v11 **Aggregation Methods:** FedAvg, FedProx, FedAsync **Datasets:** KITTI, BDD100K, NuScenes
Evaluation Goals: We analyze trade-offs between detection accuracy, resource utilization under varying resolutions, batch sizes, weather, and lighting conditions, as well as dynamic client participation, to guide robust and scalable FL deployment in CAVs.

Model	Centralised			FedAvg			FedProx			FedAsync		
Dataset	V5	V8	V11	V5	V8	V11	V5	V8	V11	V5	V8	V11
KITTI	78.4	81.6	77.6	82.9	87.5	84.0	83.4	87.9	84.1	78.3	82.0	80.1
BDD	58.9	61.4	60.2	59.3	61.5	60.5	59.6	61.5	60.7	42.3	45.7	44.1
Nuscenes	53.2	56.6	55.9	57.3	60.6	59.4	57.7	60.8	59.9	38.6	41.8	41.7

Tab 1. mAP Comparison of Centralized and Federated Learning across YOLO versions

Model	Inference (ms)		
Dataset	V5	V8	V11
KITTI	0.9	1.1	1.2
BDD	1.3	1.6	3.1

Tab 2. Inference across models

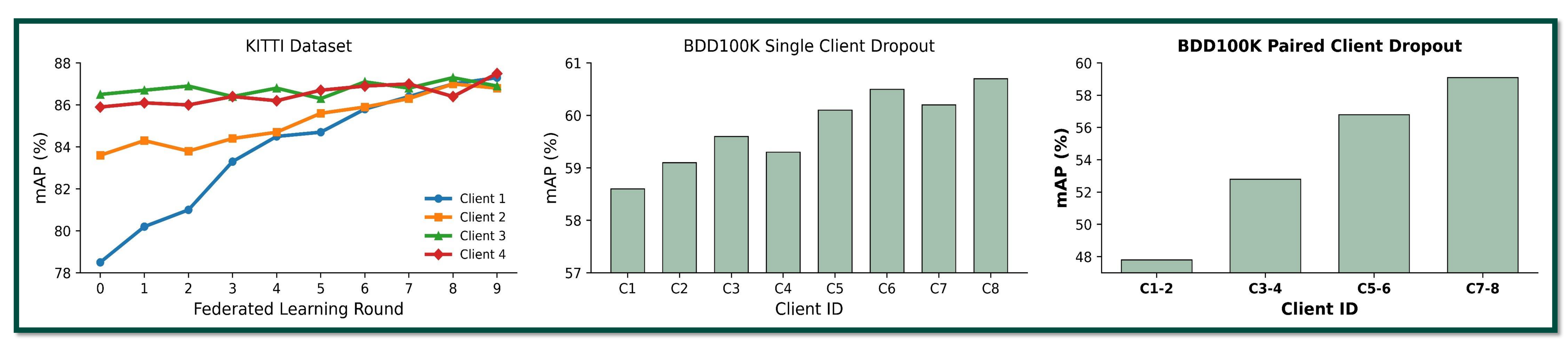

Figure 4. Effect of client dropout on mAP for KITTI and BDD100K YOLOv8

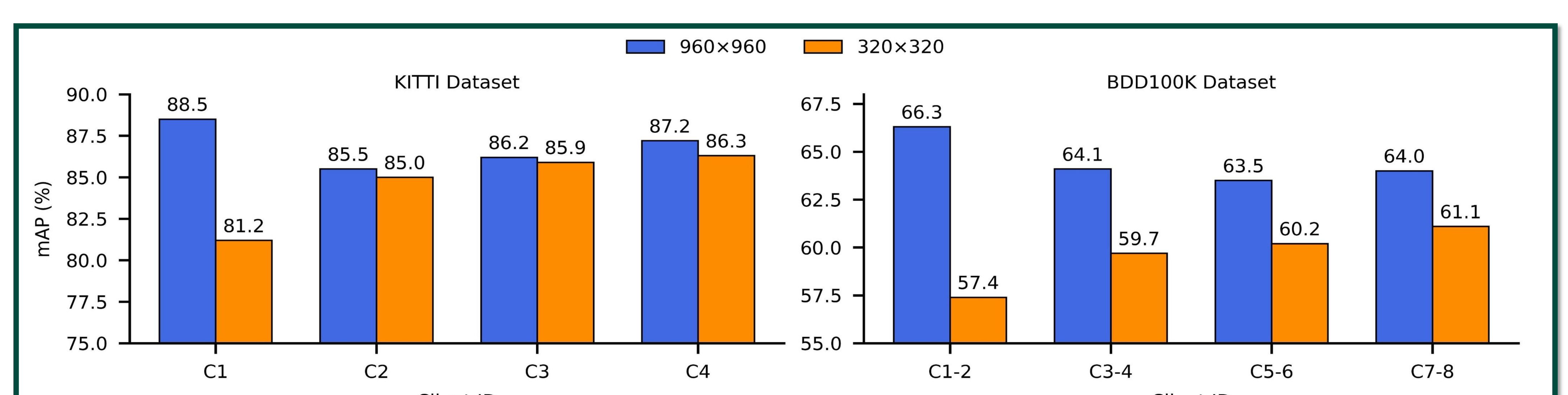

Figure 5. Client resolution heterogeneity and its effect on accuracy YOLOv8

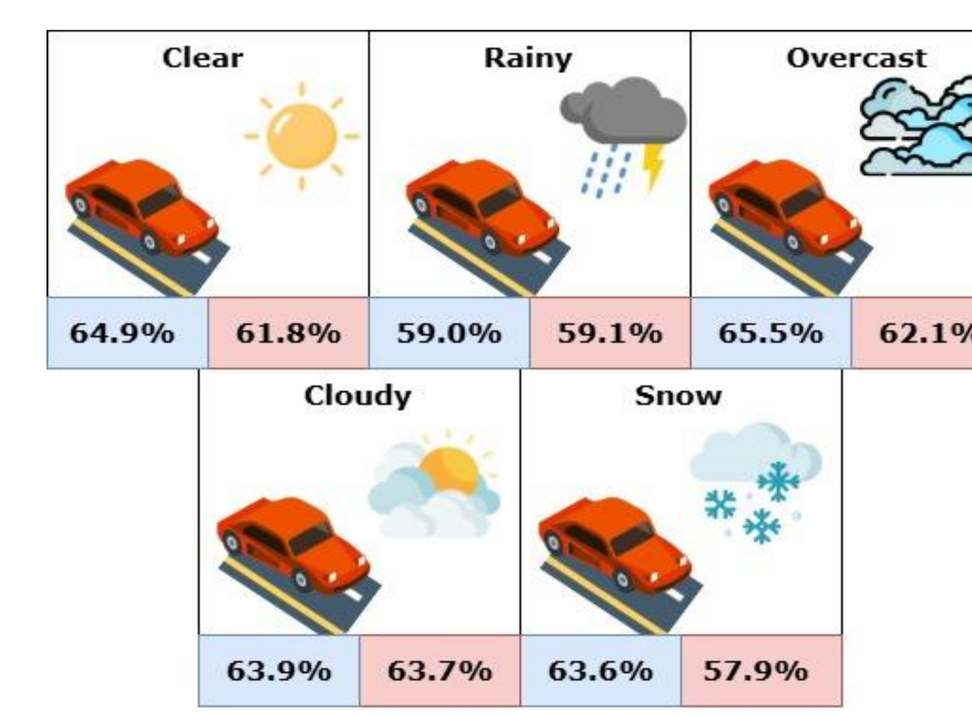

Figure 6. Weather performance. In-Domain (Blue), Cross-Domain (Red)

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